Jerome Rodrigo

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Boston, MA | Active Top Secret Clearance

Education

Northeastern University — Boston, MA
Bachelor of Science in Data Science and Mathematics — *April 2028*

Awards: Dean's List (F24, SP25, F25) — **GPA: 3.90 / 4.00**

Relevant Coursework: Graduate NLP, Object-Oriented Design, Databases, Cloud Computing, Linear Algebra, Probability and Statistics, Differential Equations, Fundamentals of Computer Science (Accelerated)

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, C++
Frameworks & Tools: FastAPI, Flask, Docker, Airflow, React, Next.js, Tailwind CSS, Git
ML & Data Science: PyTorch, Scikit-Learn, OpenCV, Hugging Face, LangChain, Matplotlib, NumPy
Databases & Cloud: PostgreSQL, MongoDB, DuckDB, pgvector, AWS, Azure, OpenShift

Experience

Regeneron — Rensselaer, NY
Machine Learning Intern — *June 2026 – Present*

- Designing and building multi-agent RAG system in LangChain with critic agent SQL validation and history traversal, enabling scientists to query drug manufacturing data in natural language
- Developing LLM-powered dashboards that surface outliers and explain process deviations for research scientists

MORSE Corp — Boston, MA
ML Software Engineer Co-op — *January 2026 – June 2026*

- Led backend optimization and architecture of the CV analytics platform, enabling on-demand model evaluation and first external client deployment and replacing manual report-based workflows
- Migrated metric ingestion to Parquet/DuckDB with column-pruned queries and memory-bounded scans, dropping analytical query time from 40s to 2s across 70 models and 35M+ rows
- Built TOPSIS scoring engine that ranks CV models across metrics, used by engineers and stakeholders to select models for deployment
- Deployed via Dockerized services, Airflow DAGs, S3-compatible storage, OpenShift, and GitLab CI/CD pipelines

Boston Children's Hospital — Boston, MA
Machine Learning Research Intern — *September 2025 – December 2025*

- Built automated organoid segmentation pipeline in OpenCV, reducing manual QC from hours to minutes for pediatric GI imaging studies
- Extracted per-organoid morphology and intensity features (area, circularity, skewness) from masks for analysis
- Identified treatment-driven response patterns across 500+ organoid samples using PCA, UMAP, and clustering

Leadership

Generate: Northeastern's Student Product Development Studio — Boston, MA
Tech Lead — *July 2025 – Present*

- Led 8-member team shipping ML and NLP systems for two startups non-technical clients
- Partnered with non-technical founders to scope and deliver full-stack ML applications across two startups
- Architected backend per client: semantic search, Docker Compose services, and enforced CI/CD coverage
- Drove planning, code reviews, and debugging to ship production ML products under real client deadlines

Khoury College of Computer Sciences — Boston, MA
Teaching Assistant - Object-Oriented Design — *August 2025 – December 2025*

- Taught weekly labs of 40+ students in an Object-Oriented Design course, covering Java, UML, and design patterns
- Supported students through office hours, helping them debug code and reinforce OOD principles

Projects

Nutrition Recommender | *HuggingFace, FastAPI, Docker, React, Python, JavaScript* — June 2025

- Built a system that delivers personalized nutrition advice by combining user biometrics with health research
- Implemented RAG pipeline with semantic chunking, React frontend, FastAPI, and Docker deployment

Blackjack RL | *Gymnasium, PyTorch, NumPy, Python* | jerodrigo.com/blackjackrl.pdf — March 2025

- Implemented a Deep Q-Learning agent in a custom Gymnasium Blackjack environment to learn optimal strategies
- Developed a paper examining the math of Deep Q-Learning, neural networks, and the reasoning for using DQNs